

HAIL, Saudi Arabia

WMO: 403940

Lat: 27.438N

Lon: 41.686E

Elev: 3331

StdP: 13.01

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.4	36.9	10.3	10.5	55.4	14.2	12.7	55.6	26.8	65.0	23.4	62.6	3.3	180	0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	27.9	107.7	65.7	106.1	65.0	104.4	64.4	67.5	101.8	66.5	101.0	65.6	100.2	9.0	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.3	85.3	68.7	57.2	79.1	67.9	55.1	73.1	68.3	33.8	101.5	32.9	101.3	32.2	99.5	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.2	20.9	17.9	25.9	110.2	4.2	1.4	22.8	111.3	20.4	112.1	18.0	112.9	14.9	114.0		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			22.7	71.6	4.1	3.2	19.7	73.9	17.3	75.8	15.0	77.6	12.0	79.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	73.2	52.4	56.4	63.2	72.6	81.8	87.9	89.9	91.1	86.4	76.9	63.2	55.0		
	DBStd	14.87	6.42	6.60	6.68	5.81	4.85	3.42	3.77	3.61	3.97	4.76	6.29	5.93		
	HDD50	86	47	15	1	0	0	0	0	0	0	0	2	21		
	HDD65	1178	391	249	114	9	0	0	0	0	0	1	104	310		
	CDD50	8539	123	194	411	679	985	1138	1235	1274	1091	833	399	177		
	CDD65	4156	1	8	60	238	520	688	770	809	641	368	52	1		
	CDH74	65981	38	162	852	2818	7366	11509	13460	14143	10163	4751	654	65		
Wind	WSAvg	42188	2	23	272	1236	4275	7708	9366	9997	6637	2500	170	2		
	WSAvg	7.5	7.2	7.9	8.3	8.6	8.5	7.8	7.6	6.8	6.5	6.9	6.6	6.7		
Precipitation	PrecAvg	3.3	0.5	0.2	0.9	0.4	0.3	0.0	0.0	0.0	0.0	0.3	0.5	0.3		
	PrecMax	9.6	1.6	1.3	3.3	1.9	1.6	0.1	0.2	0.0	0.0	1.2	1.8	2.2		
	PrecMin	0.4	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	1.8	0.4	0.4	1.0	0.5	0.4	0.0	0.0	0.0	0.0	0.4	0.6	0.5		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.6	83.5	91.6	97.0	103.8	107.5	109.5	109.6	106.2	98.6	87.7	79.1		
		MCWB	55.4	57.1	58.7	60.8	63.6	64.6	65.9	66.9	65.0	62.7	59.7	55.0		
	2%	DB	73.8	78.8	86.5	93.5	100.7	105.6	107.6	107.8	104.3	96.5	84.6	75.5		
		MCWB	53.4	55.7	57.5	61.0	62.8	64.1	65.5	66.1	64.4	61.6	57.8	55.4		
	5%	DB	70.1	75.1	82.5	91.0	98.5	103.9	105.8	106.4	102.5	94.6	81.5	72.0		
		MCWB	52.2	53.5	56.3	59.8	62.3	63.7	64.7	65.6	63.8	60.8	57.2	53.8		
	10%	DB	66.3	71.4	78.7	87.5	96.2	102.0	103.9	105.4	100.6	91.8	77.4	68.3		
		MCWB	50.3	51.8	54.8	58.8	61.6	62.8	64.1	65.2	63.1	59.9	56.0	52.1		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.3	60.4	62.8	66.0	67.7	68.1	68.2	69.7	67.3	66.0	63.9	61.4		
		MCDB	72.1	75.3	79.7	82.1	92.2	100.9	104.7	104.1	101.9	86.0	75.4	71.9		
	2%	WB	56.4	57.4	59.8	63.7	65.6	65.7	66.6	67.8	65.7	63.7	61.9	58.1		
		MCDB	68.9	73.0	79.5	83.8	91.6	101.9	104.0	104.6	101.0	86.8	74.4	70.2		
	5%	WB	53.9	55.3	57.9	62.1	64.0	64.5	65.6	66.6	64.7	62.3	60.2	55.9		
		MCDB	66.0	71.4	77.7	82.6	91.7	101.3	103.1	103.5	100.4	87.4	72.9	67.8		
	10%	WB	51.6	53.1	56.0	60.6	62.8	63.4	64.6	65.8	63.7	61.1	58.5	53.6		
		MCDB	63.7	68.9	75.7	81.9	90.8	99.7	102.0	102.7	98.8	87.4	71.7	65.2		
Mean Daily Temperature Range	5% DB	MDBR	24.0	25.1	25.5	25.8	26.0	27.7	27.1	27.9	28.7	28.3	24.1	24.2		
		MCDBR	28.4	29.4	30.1	29.1	28.4	29.3	28.6	28.6	29.4	30.2	29.4	28.7		
		MCWBR	14.8	15.1	14.0	11.8	11.3	11.6	11.3	11.3	11.7	12.8	13.3	15.0		
	5% WB	MCDBR	24.5	25.9	26.8	25.6	26.5	27.6	27.2	27.2	27.9	27.4	22.9	24.3		
		MCWBR	14.1	14.2	13.6	11.4	10.9	11.7	11.3	11.4	11.6	11.8	11.3	13.6		
Clear-Sky Solar Irradiance	taub		0.341	0.383	0.447	0.521	0.543	0.492	0.469	0.510	0.476	0.468	0.416	0.365		
	taud		2.254	2.105	1.909	1.763	1.727	1.844	1.937	1.827	1.884	1.894	2.026	2.170		
	Ebn at Noon		284	280	267	251	245	257	262	251	255	247	253	268		
	Edh at Noon		37	47	61	72	75	66	60	67	61	57	47	39		
All-Sky Solar Radiation	RadAvg		1335	1625	1964	2180	2364	2608	2538	2352	2085	1727	1354	1225		
	RadStd		75	70	68	67	98	62	71	64	64	57	87	54		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.88	N/A	N/A	+0.43	N/A	N/A	N/A	-94	+387	+297
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page